The Titanic Dataset: Predicting Survival Using Machine Learning

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Abstract

The goal of this project is to determine whether passengers of the famous Titanic cruise would have survived, given several factors about each individual passenger. The dataset we are using is the famous Titanic dataset from Kaggle, which contains 891 passengers, whether they survived, and 11 other variables. These variables are passenger ID, their ticket class, their name, gender, age, number of siblings or spouses aboard the Titanic, number of parents or children aboard the Titanic, their ticket number, passenger fare, cabin number, and which port they embarked from. We will first use exploratory data analysis to determine if there are similarities in the passengers that survived. Next, for us to determine which passengers would have survived depending on those variables, we will use three machine learning techniques - Logistic Regression, Decision Tree, and K Nearest Neighbors. To ensure the most accurate model, we will first use techniques such as standardization and label encoding to transform the dataset. Finally, we will compare the performance of these three models and determine which is the most effective in predicting passenger survival.

Introduction

As one of the most infamous incidents in history, the sinking of the Titanic has been often-referenced in day-to-day life and culture. A largely discussed topic surrounding the sinking has been whether those who survived had done so purely due to random chance. During the evacuation, women and children were prioritized in the queue for lifeboats; this could lead to the majority of the survivors being those groups. We will use the dataset and its variables to explore if we can accurately predict survival using related factors such as gender or age.

Related Work

This project draws its inspiration from many of the past attempts to determine passenger survival rates using the Titanic dataset. Ekinel et. al (2018) explores comparisons between machine learning techniques such as the ones used in this project using Titanic data. A similar study was done by Kakde & Agrawal (2018) that used a combination of exploratory data analytics and machine learning techniques.

Methodology

By loading the dataset, we see from Table 1 that there are 891 passengers, and their passenger ID, survival status, class, name, sex, age, number of siblings, number of parents and children, ticket number, fare price, which cabin they are, and which port they embarked from. We also see from Table 1 that the cabin column contains over 700 null values, so we choose to drop that column. There are also over 100 missing values in the age column, and after examining the distribution in the histogram in Figure 1, we impute the column with the median age. Lastly, we impute the 2 missing values in the embarked column with the most frequently appearing port, S.

Table 1

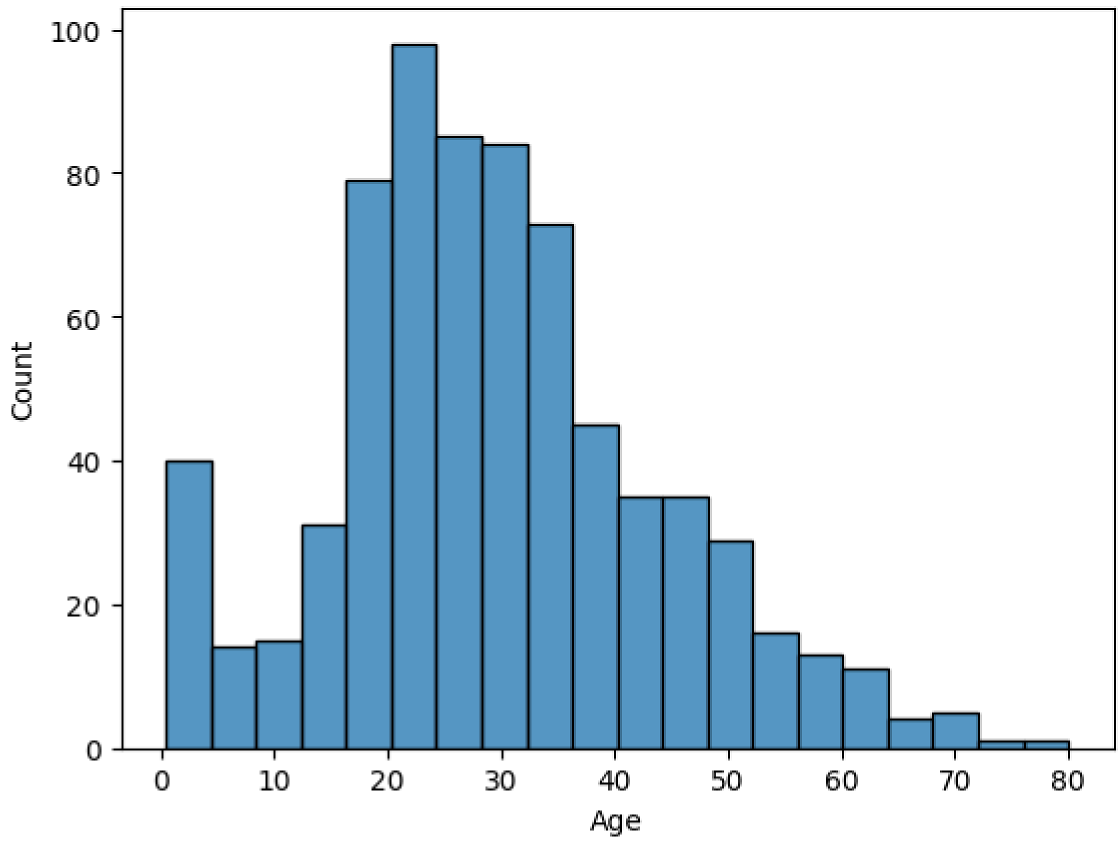
Titanic Dataset Information

Column	**Non-Null Count**	**Data Type**
PassengerId	891	Numeric
Survived	891	Categorical
Pclass	891	Categorical
Name	891	Text
Sex	891	Categorical

Age	714	Numeric
SibSp	891	Categorical
Parch	891	Categorical
Ticket	891	Text
Fare	891	Numeric
Cabin	204	Text
Embarked	889	Categorical

Figure 1

Distribution of Age



After imputing, we remove the unnecessary columns passenger ID, name, and ticket for modelling from our dataset, and encode the text categorical variables sex and port embarked to numeric. Table 2 shows the summary statistics for the remaining columns.

Table 2

Summary Statistics of Titanic Dataset after Preprocessing

	Survived	**Pclass**	**Sex**	**Age**	**SibSp**	**Parch**	**Fare**	**Embarked**
count	891.00	891.00	891.00	891.00	891.00	891.00	891.00	891.00
mean	0.38	2.31	0.65	29.36	0.52	0.38	32.20	1.54
std	0.49	0.84	0.48	13.02	1.10	0.81	49.69	0.79
min	0.00	1.00	0.00	0.42	0.00	0.00	0.00	0.00
25%	0.00	2.00	0.00	22.00	0.00	0.00	7.91	1.00
50%	0.00	3.00	1.00	28.00	0.00	0.00	14.45	2.00
75%	1.00	3.00	1.00	35.00	1.00	0.00	31.00	2.00
max	1.00	3.00	1.00	80.00	8.00	6.00	512.33	2.00

To examine which types of passengers were more likely to survive, we perform some exploratory data analysis. Figure 2 shows the pair plot for each remaining factor in the dataset. We see that it is difficult to determine any relationships from this output, so we will view the correlations for each factor as well.

Figure 2

Pair Plots for Titanic Dataset Relationships

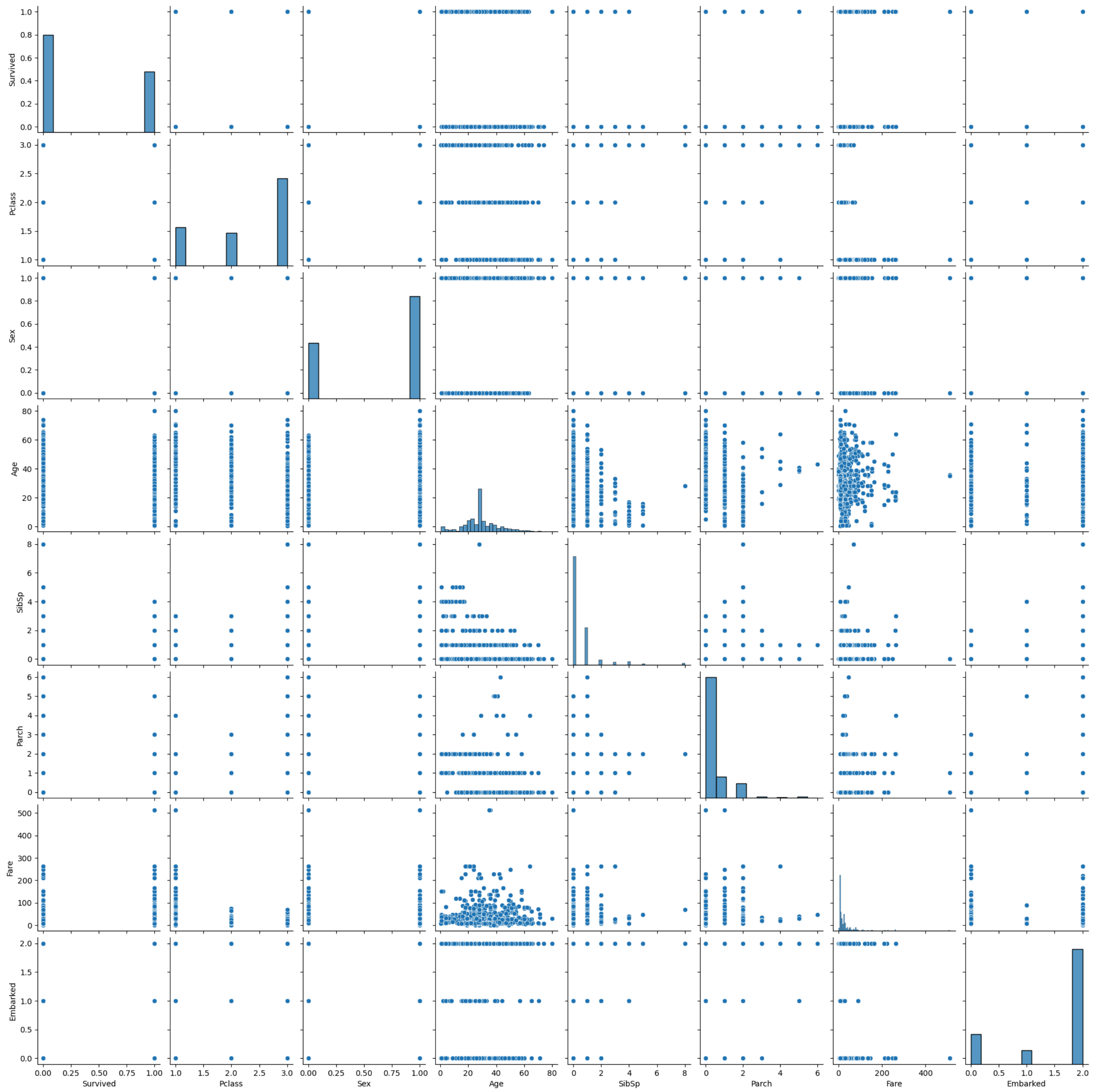


Table 3 and Figure 3 examines the correlations between each variable through a correlation table and a heatmap respectively. We see that the odds of survival are most significantly correlated with class, sex, and fare. To confirm this, we see in Table 4, the median of each variable by whether the passenger survived or not, those who survived were of a higher class, female, and had higher fare.

Table 3

Titanic Dataset Correlation Table

	Survived	**Pclass**	**Sex**	**Age**	**SibSp**	**Parch**	**Fare**	**Embarked**
Survived	1.00	-0.34	-0.54	-0.06	-0.04	0.08	0.26	-0.17
Pclass	-0.34	1.00	0.13	-0.34	0.08	0.02	-0.55	0.16
Sex	-0.54	0.13	1.00	0.08	-0.11	-0.25	-0.18	0.11
Age	-0.06	-0.34	0.08	1.00	-0.23	-0.17	0.10	-0.02
SibSp	-0.04	0.08	-0.11	-0.23	1.00	0.41	0.16	0.07
Parch	0.08	0.02	-0.25	-0.17	0.41	1.00	0.22	0.04
Fare	0.26	-0.55	-0.18	0.10	0.16	0.22	1.00	-0.22
Embarked	-0.17	0.16	0.11	-0.02	0.07	0.04	-0.22	1.00

Figure 3

Titanic Dataset Correlation Heatmap

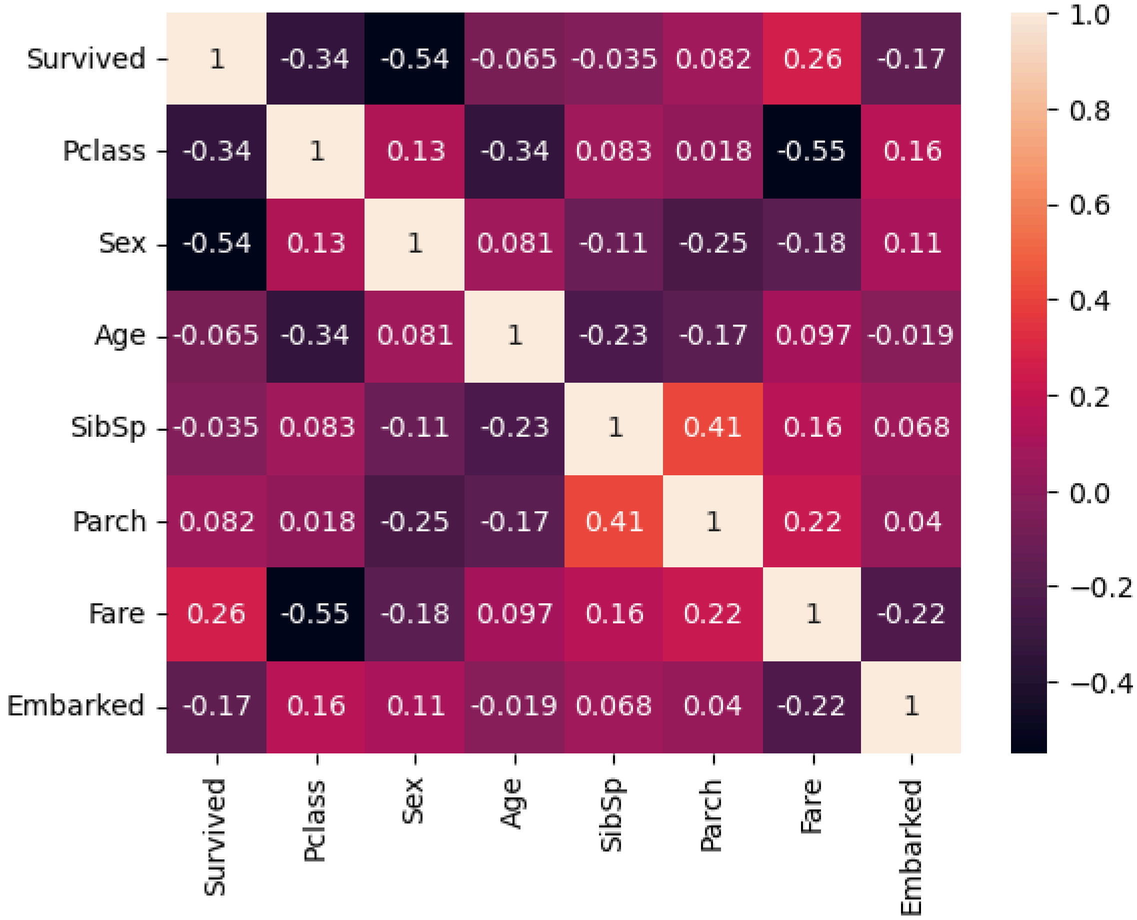


Table 4

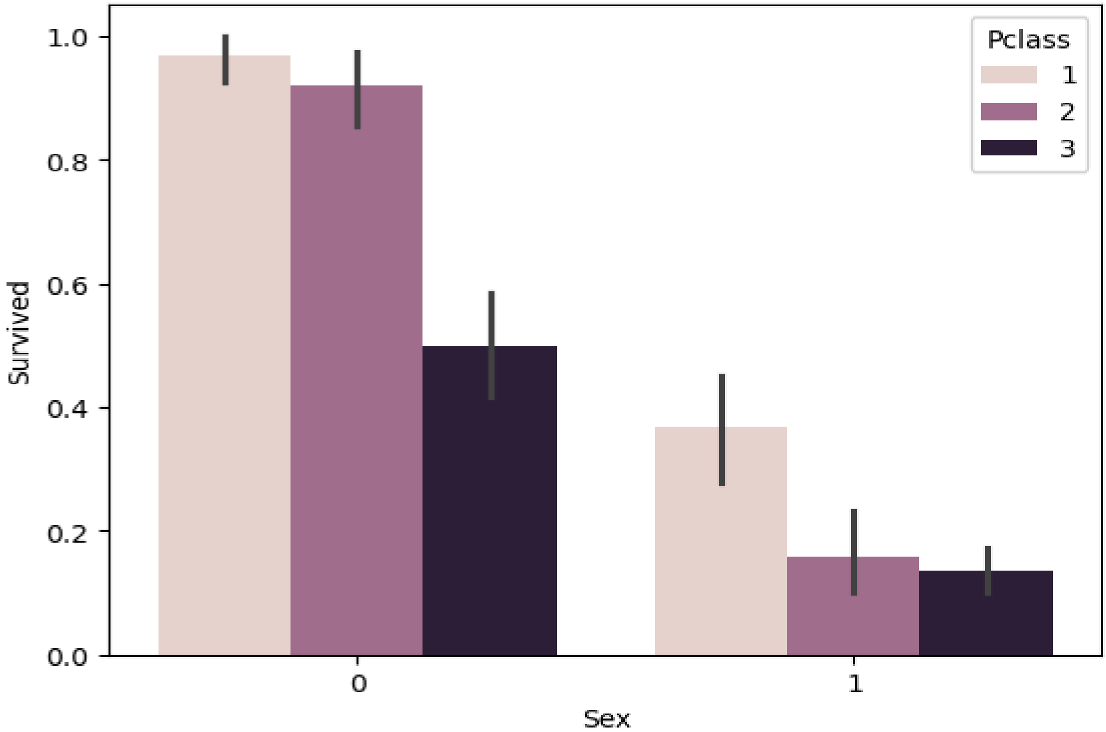
Medians Grouped By Survival Status

Survived	**Pclass**	**Sex**	**Age**	**SibSp**	**Parch**	**Fare**	**Embarked**
0	3.0	1.0	28.0	0.0	0.0	10.5	2.0
1	2.0	0.0	28.0	0.0	0.0	26.0	2.0

Finally, we can also confirm from Figure 4 that females were more likely to survive, as well as those of a higher class.

Figure 4

Barplot of Survival Probability by Sex and Class



After exploratory data analysis to determine similarities in surviving passengers, we will now perform Logistic Regression, Decision Tree, and K Nearest Neighbors to determine which model is the most effective in classifying survival status using the variables provided. Since there are significantly more passengers that did not survive than those who did, there is a class imbalance in the dataset. A class imbalance can lead to overfitting on the more present class, so we combat this problem by using stratified sampling to split our dataset into training and testing sets. We use a random seed of 42 for splitting and all of our models to ensure reproducibility. Also, for K Nearest Neighbors, we use Grid Search to determine the best number of neighbors.

The final step before evaluating our models' accuracies is to standardize the data. Since we are splitting our dataset into a training and testing set, we first standardize the training set, and then standardize the testing set using the training set. This ensures that the testing set is still unseen data to test our models on.

Results

We can see from Table 5 that K Nearest Neighbors is the most accurate model for classifying survival. Also from Tables 6, 7, and 8, we can view the confusion matrices for each model. These matrices confirm that KNN has the most true positives and negatives and Decision Tree the least.

Table 5

Accuracy Comparisons of Models

Model	**Accuracy**
KNN	0.7803
Logistic Regression	0.7713
Decision Tree	0.7309

Table 6

Confusion Matrix of K Nearest Neighbors

	Survived	**Did Not Survive**
Survived	119	18
Did Not Survive	31	55

Table 7

Confusion Matrix of Logistic Regression

	Survived	**Did Not Survive**
Survived	114	23
Did Not Survive	28	58

Table 8

Confusion Matrix of Decision Tree

	Survived	**Did Not Survive**
Survived	108	29
Did Not Survive	31	55

Discussion

After exploratory data analysis, we determine that the passengers most likely to survive are of a higher class, female, and with higher fare. This makes sense as women and children were the first on the lifeboats off of the Titanic when it began sinking. Those with higher fare also were logically of the higher class, which were also evacuated first. It is surprising that age was not as significant of a factor in survival status, as, as previously stated, children were among the first off the ship.

We also determine from the accuracies of each model that K Nearest Neighbors with Grid Search is the most accurate model in predicting survival status of passengers. This could be because we did not use ensemble models like Random Forest for our other selected models. Overall, the accuracies for each model are still very low. Ways to increase accuracy include performing feature selection through PCA or being more accurate in imputing age, such as grouping medians by class or other variables.

Conclusion

Summary of the key findings and suggestions for future research.

References

Ekinci, E., Omurca, S. İ., & Acun, N. (2018, December). A comparative study on machine learning techniques using Titanic dataset. In *7th international conference on advanced technologies* (pp. 411-416).

Kakde, Y., & Agrawal, S. (2018). Predicting survival on titanic by applying exploratory data analytics and machine learning techniques. *International Journal of Computer Applications*, *179*(44), 32-38.

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